# 1.2.72

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# Question

Evaluate the integral:

$$\int_0^3 \int_0^{3-x} (6 - x - y)\, dy\, dx \tag{1}$$

**Options:**

(a) 13.5

(b) 18.0

(c) 40.5

(d) 54.0

# Step 1: Inner Integral

Evaluate the inner integral:

$$\int_0^{3-x} (6 - x - y)\, dy \tag{2}$$

Split the integral:

$$= \int_0^{3-x} (6 - x)\, dy - \int_0^{3-x} y\, dy \tag{3}$$

$$= (6 - x)(3 - x) - \frac{(3 - x)^2}{2} \tag{4}$$

# Step 2: Simplification

$$(6-x)(3-x) = 18 - 9x + x^2 \tag{5}$$

$$\frac{(3-x)^2}{2} = \frac{9 - 6x + x^2}{2} \tag{6}$$

$$\Rightarrow \int_0^{3-x} (6 - x - y)\, dy = 18 - 9x + x^2 - \frac{9 - 6x + x^2}{2} \tag{7}$$

$$= \frac{27 - 12x + x^2}{2} \tag{8}$$

# Step 3: Outer Integral

$$\int_0^3 \frac{27 - 12x + x^2}{2}\, dx = \frac{1}{2}\int_0^3 (27 - 12x + x^2)\, dx \tag{9}$$

$$= \frac{1}{2}\left[27x - 6x^2 + \frac{x^3}{3}\right]_0^3 \tag{10}$$

# Step 4: Final Calculation

Substitute limits:

$$= \frac{1}{2}(81 - 54 + 9) \tag{11}$$

$$= \frac{1}{2} \times 36 = 18 \tag{12}$$

# Region of Integration

**In one line:** Volume under the plane $z = 6 - x - y$ over a triangular region in the $xy$-plane.

# Python Code

```
    import numpy as np
import matplotlib.pyplot as plt

fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

# --- Top surface (plane) ---
x = np.linspace(0, 3, 100)
y = np.linspace(0, 3, 100)
X, Y = np.meshgrid(x, y)

Z = 6 - X - Y
mask = (X + Y <= 3)
Z = np.where(mask, Z, np.nan)

ax.plot_surface(X, Y, Z, alpha=0.7)
```

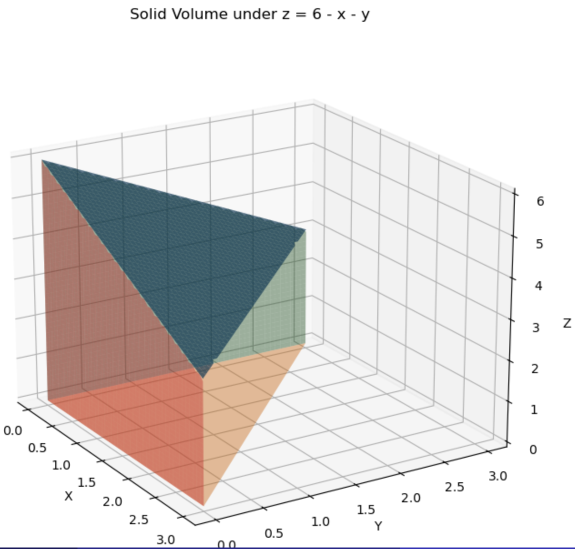
# Python Code

```
# --- Bottom (triangle) ---
ax.plot_trisurf(
    [0, 3, 0],
    [0, 0, 3],
    [0, 0, 0],
    alpha=0.4
)

# --- Side 1 (x = 0 plane) ---
y_side = np.linspace(0, 3, 50)
z_side = np.linspace(0, 6, 50)
Y_side, Z_side = np.meshgrid(y_side, z_side)
X_side = np.zeros_like(Y_side)

# Keep only valid region
mask_side = (Z_side <= 6 - Y_side)
Z_side = np.where(mask_side, Z_side, np.nan)

ax.plot_surface(X_side, Y_side, Z_side, alpha=0.4)
```

# Python Code

```python
# --- Side 2 (y = 0 plane) ---
x_side = np.linspace(0, 3, 50)
z_side = np.linspace(0, 6, 50)
X_side, Z_side = np.meshgrid(x_side, z_side)
Y_side = np.zeros_like(X_side)

mask_side = (Z_side <= 6 - X_side)
Z_side = np.where(mask_side, Z_side, np.nan)

ax.plot_surface(X_side, Y_side, Z_side, alpha=0.4)

# Labels
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')

ax.set_title('Solid Volume under z = 6 - x - y')

plt.show()
```

# Plot



Solid Volume under z = 6 - x - y
X
Y
Z
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
1
2
3
4
5
6